KUMASI TECHNICAL UNIVERSITY

COMPUTER SCIENCE DEPARTMENT

BACK – END WEB DEVELOPMENT TECHNOLOGY

COURSE CODE: BCT 205

MERCY VICENTIA ADU GYAMFI (MRS)

PHP BASICS

Overview of PHP

- **Definition:**
  - PHP stands for **Hypertext Preprocessor**.
  - It is a **server-side scripting language** used to create **dynamic web pages**.
- **Key Features:**
  - **Cross-Platform:** Works on Windows, Linux, macOS.
  - **Embedded in HTML:** PHP code can be placed inside HTML.
  - **Supports Databases:** Works well with MySQL, PostgreSQL, SQLite, etc.
  - **Fast & Efficient:** Handles dynamic content quickly.
- **Common Uses:**
  - Form processing (login/register)
  - Content management systems (e.g., WordPress)
  - E-commerce websites
  - Dynamic web applications

Setting Up a PHP Page

Steps to Create a PHP Page:

- Ensure your server is running (**e.g., XAMPP or WAMP).**
- Launch Apache **(for PHP) and MySQL** (for database).
- Ensure the server is running before testing **PHP files**.
- Use a text editor or IDE (e.g., VS Code, Bracket, PhpStorm).
- Save file with a **.php** extension.
- Write PHP code inside **<?php ?>** tags.

Write Basic PHP Code

```
<?php
echo "Hello, World!";
?>
```

Place File in Server Directory

- XAMPP: htdocs folder
- WAMP: www folder
- This makes the file accessible via a browser

Run in Browser

- Open browser and navigate to: http://localhost/filename.php

VARIABLES IN PHP

Definition:

- A **variable** is a container used to **store data** such as text, numbers, or other values.

Rules:

- Variables must **start with $** sign.
- Should start with a letter or underscore (_).
- Are Case-sensitive ($name ≠ $Name)
- Can contain letters, numbers, and underscores
- Cannot start with a number
- No spaces allowed

Example :

```
$name = "Mercy"; //string variable

$age = 20; // integer variable
```

Data Types in PHP:

- String – Text data

Example: $name = "Mercy";

- Integer – Whole numbers

Example: $age = 20;

- Float – Decimal numbers

Example: $price = 99.99;

- Boolean – True or False

Example:$isStudent = true;

- Array – Multiple values

Example: $colors = array("Red", "Blue", "Green");

VARIABLES Example

Example:

```
<?php
$firstName = "Mercy";
$age = 20;
echo "Name: $firstName, Age: $age";
?>
```

Numbers in PHP

Definition:

- Numbers in PHP are used to perform **mathematical operations**.

Types of Numbers in PHP:

- Integer and Float / Double

Basic Arithmetic Operators:

Operator	Description	Example
+	Addition	$a + $b
-	Subtraction	$a - $b
*	Multiplication	$a * $b
/	Division	$a / $b
%	Modulus (remainder)	$a % $b

Numbers in PHP

Mathematical Functions:

- abs($x) → Absolute value
- round($x) → Round to nearest I
- ntegerceil($x) → Round up
- floor($x) → Round down
- sqrt($x) → Square root
- pow($x, $y) → Power function

Example Code:

```
<?php
$a = 10;
$b = 3;
echo $a + $b; // 13
echo $a % $b; // 1
echo sqrt(16); // 4
?>
```

Example:

```
$sum = 10 + 20;
echo $sum; // Output: 30
```

Strings in PHP

Definition:

- A **string** is a sequence of characters used to store text data.

Creating Strings:

- **Strings** can be created using **single quotes** or **double quotes**.

Example:

```
$name = "Mercy";

$course = 'Web Programming’;

Echo $name;// output: Mercy
```

Strings in PHP

Difference Between Quotes:

- Double quotes (" ") interpret variables.
- Single quotes (' ') treat content as plain text.

String Operations:

- **Concatenation:** Joining strings using the dot (.) operator

Example:

```
$fullName = "Mercy" . " Adu";
```

Common String Functions:

strlen() – Length of a string

strtoupper() – Convert to uppercase

strtolower() – Convert to lowercase

substr() – Extract part of a string

strpos() – Find position of a word

Strings in PHP

Example:

```
<?php
$text = "Hello World";
echo strlen($text);
echo strtoupper($text);
?>
```

Uses of Strings:

- Displaying messages
- Handling form input
- Storing text data

Logical and Comparison Operators in PHP

Comparison Operators:

- Used to compare two values
- Return true or false

Operator	Meaning	Example
==	Equal to	$a == $b
===	Identical (value & type)	$a === $b
!=	Not equal to	$a != $b
>	Greater than	$a > $b
<	Less than	$a < $b
>=	Greater than or equal to	$a >= $b
<=	Less than or equal to	$a <= $b

Logical and Comparison Operators in PHP

Logical Operators:

- **Used to perform logical operations in conditions.**

Operators:

- && (AND): Both conditions must be true.
- || (OR): At least one condition must be true.
- ! (NOT): Negates a condition.

Example:

```php
<?php
$a = 10;
$b = 5;
if ($a > $b && $a != 0) {
    echo "Condition is true";
}
?>
```

Logical and Comparison Operators in PHP

Uses:

- Decision making
- Form validation
- User authentication

Example:

```
$age = 20;
if ($age >= 18 && $age <= 30) {
    echo "You are eligible.";
}
```

Arrays in PHP

Definition:

- An **array** is a data structure used to **store multiple values** in a single variable.

Types of Arrays in PHP:

1. **Indexed Arrays**
   - An **indexed array** is an array where elements are stored using **numeric indexes**, starting from **0**.

Creating an Indexed Array: $colors = array("Red", "Blue", "Green"); **Or**

$colors = ["Red", "Blue", "Green"];

Accessing Array Elements: echo $colors[0]; // Red echo $colors[1]; // Blue

Modifying Array Elements: $colors[1] = "Yellow";

Adding Elements: $colors[] = "Black";

Looping Through Indexed Arrays:

```
foreach ($colors as $color) {
    echo $color;
}
```

Arrays in PHP

Types of Arrays in PHP:

1. **Indexed Arrays**

- **Common Functions:**
  - count() – Returns number of elements
  - sort() – Sorts array in ascending order
- **Uses:**
  - Storing lists (names, courses, items)
  - Displaying data in loops

Arrays in PHP

Types of Arrays in PHP:

2. Associative Arrays in PHP

Definition:

- An **associative array** uses **named keys** instead of numeric indexes to store values.

Creating an Associative Array:

```
$student = array(
  "name" => "Mercy",
  "age" => 20,
  "course" => "Web Programming"
);
echo $student["name"];
echo $student["age"];
```

Arrays in PHP

Types of Arrays in PHP:

2. Associative Arrays in PHP

- **Common Functions:**
  - count() – Number of elements
  - array_keys() – Returns all keys
  - array_values() – Returns all values
- **Uses:**
  - Storing records (student details, user profiles)
  - Managing database records

Arrays in PHP

Types of Arrays in PHP:

3. Multidimensional Arrays in PHP

Definition:

- A **multidimensional array** is an array that **contains one or more arrays**.

Purpose:

- Used to store **complex data structures** such as tables, matrices, and database records.

Arrays in PHP

Types of Arrays in PHP:

3. Multidimensional Arrays in PHP

Creating a Multidimensional Array:

```
$students = array(
  array("Mercy", 20, "IT"),
  array("John", 21, "CS"),
  array("Ama", 19, "BTech")
);
```

Accessing Elements:

```
echo $students[0][0]; // Mercy
echo $students[1][2]; // CS
```

Arrays in PHP

3. Multidimensional Arrays in PHP

Looping Through Multidimensional Arrays:

```
foreach ($students as $student) {
  foreach ($student as $value) {
    echo $value . " ";
  }
}
```

Real-Life Examples:

- Student records
- Examination results
- Product lists

Advantages:

- Organizes complex data
- Efficient data handling

Control Structures in PHP

Definition:

- Control structures are used to **control the flow of program execution** based on conditions or repetitions.

1. **Conditional Statements**

Used to make **decisions** in a program.

a) if Statement

```
if ($age >= 18) {
    echo "Eligible";
}
```

Control Structures in PHP

b) **if–else Statement**

```
if ($score >= 50) {
    echo "Pass";
} else {
    echo "Fail";
}
```

c) switch Statement

```
switch ($day) {
    case "Monday":
        echo "Start of week";
        break;
    default:
        echo "Another day";
}
```

Control Structures in PHP

2. Looping Structures

Used to **repeat actions**.

a) for Loop

```
for ($i = 1; $i <= 5; $i++) {
    echo $i;
}
```

b) while Loop

```
$i = 1;
while ($i <= 5) {
    echo $i;
    $i++;
}
```

Control Structures in PHP

c) foreach Loop

```
foreach ($colors as $color) {
    echo $color;
}
```

Uses of Control Structures

- Decision making
- Repeating tasks
- Validating user input
- Processing arrays and data

Questions?

1. Why is PHP considered a popular backend programming language?

A) It is difficult to learn but very powerful

B) It is open-source, flexible, and integrates easily with various front-end technologies and databases

C) It is primarily used for mobile app development

D) It only supports MySQL databases

2. What is the primary role of backend technologies in web development?

A) To design the user interface of the website

B) To manage and process data and ensure efficient communication between the front-end and the server

C) To handle only the visual elements of a website

D) To optimize the website's search engine ranking

3. Which of the following is NOT an advantage of using MySQL as a database management system?

A) MySQL is highly scalable and can handle large data sets efficiently

B) MySQL provides robust security features to protect user data

C) MySQL is not compatible with PHP

D) MySQL is one of the most trusted relational database systems in the industry

4. Which tool can be used to manage MySQL databases in a web browser?

A) MySQL Workbench

B) phpMyAdmin

C) Visual Studio Code

D) Docker

Assignment

Practical Implementation

Write a PHP program that creates a multidimensional array to store the following student

Name	Age	Programme
Ama	20	BTech IT
Kojo	22	BTech CS

Your program should:

- Store the data in a multidimensional array
- Display each student's details on a new line

THANK YOU !

NO MATTER WHICH FIELD OF WORK YOU WANT TO GO IN, IT IS OF GREAT IMPORTANCE TO LEARN AT LEAST ONE PROGRAMMING LANGUAGE.